

OPERATIONAL IMPACT

Visible ROI in critical rotating machinery

We turn vibration into specific effective decisions: fewer stoppages, lower risk and more available operating and production time.

12 kHz

High-resolution sensing

ISO

Standardized severity

- Agnostic to sensing platform and manufacturer: QARTIA's own solution or integration with the customer's sensors.
- Continuous 24/7 diagnosis of severity and rotating defect type.
- Failure prediction to schedule preventive maintenance before breakdown.



HIGH-PERFORMANCE PREDICTIVE MAINTENANCE

INDUSTRIAL INTELLIGENCE

Higher production, fewer stops

QARTIA integrated industrial solution:

- Sensing
- Signal processing
- Private connectivity with full factory coverage
- Expert diagnosis
- Predictive AI.



QARTIA PROPOSAL



From raw data to maintenance action

1) Intelligent condition capture

Time- and frequency-domain measurements to detect real changes in machine condition.

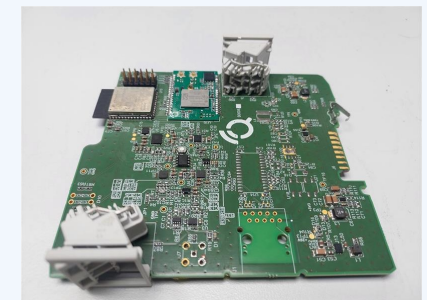
2) Real-time expert diagnosis

Severity according to ISO 10816 / 20816 family and defect classification: misalignment, unbalance, bearing wear, lack of lubrication, cavitation, etc.

3) Applied AI prediction

QARTIA-designed models and algorithms to anticipate machine degradation and plan safe shutdowns with sufficient lead time.

Request a premium plant pilot: measurable technical and economic value in weeks.



TECHNICAL ARCHITECTURE

Edge + private network + analytics

Flexible deployment: customer-side software platform (on-premises) or QARTIA-Apliquem managed cloud.

Proprietary hardware

Accelerometers up to 12 kHz, acquisition/digitization platform, and FFT for continuous measurement.

DECT NR+ communications

5G, operator-free, non-cellular private network in license-exempt spectrum, decentralized and mesh-based for coverage where other technologies perform poorly.

Expert vibration diagnosis engine

Customer-side or cloud LLM diagnostic assistant, with criteria equivalent to a Category II vibration expert.



BUSINESS VALUE

Technically, operationally and economically grounded decisions

Unifies operations, maintenance and engineering around a shared view of each asset's real risk.

MTBF

Less frequent failures

OEE

More plant availability

Failure risk by asset

Severity + frequency to prioritize where a stop costs the most.

Economic intervention priority

Actions ranked by avoided cost and production criticality.

Operations traceability

Technical reporting for maintenance, operations and management.

Turn QARTIA predictive maintenance into an industrial competitive advantage.

FAST ACTIVATION



Executive pilot in 4 phases

Phase 1 - Asset selection

Operational criticality, failure history and maintenance window.

Phase 2 - Sensor and network deployment

Field installation with coverage and data-quality validation.

Phase 3 - Diagnosis, prediction and impact report

Actionable alerts and prioritized technical recommendations. Availability KPIs, avoided risk and annualized potential savings.

Phase 4 - Commercial close

Final economic proposal, implementation plan and service agreement.

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